SUPERCONDUCTING ALUMINUM CPW $\lambda/4$ RESONATORS ON DIAMOND, 4H-SiC AND 6H-SiC SUBSTRATES FOR DIELECTRIC LOSS CHARACTERIZATION

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August 2026

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1 Introduction

Quantum computing based on superconducting circuits has emerged as one of the most promising platforms for practical quantum information processing. These architectures rely on superconducting qubits which, unlike classical transistors, leverage quantum superposition to exist in states beyond binary 0 and 1. A primary bottleneck limiting the performance of these circuits is decoherence, which is intrinsically tied to energy dissipation and quantified by the loaded quality factor (Q_L). Despite significant advancements in design and fabrication, experimental evidence suggests an upper bound on Q_L that persists across varying circuit geometries and fabrication processes [1].

While the exact mechanisms behind this limit remain an active area of research, several primary loss channels have been identified. These include non-equilibrium quasiparticle generation, cosmic radiation impacts, and dielectric losses originating from two-level systems (TLS) located at material interfaces; particularly at the substrate-metal interface [1].

Traditionally, superconducting Al quantum circuits are fabricated on silicon or sapphire substrates. In the context of bolometer research, diamond has been reported to exhibit favorable dielectric properties [2] possibly superior to those of the traditional substrates, while silicon carbide has been suggested as a comparable candidate by researchers in the field.

Building upon the challenges outlined by [1], this thesis presents a targeted investigation into the performance limits imposed by substrate-metal interfaces in two-dimensional superconducting circuits. The primary objective of this work is to design a representative CPW resonator architecture using thin-film aluminum (Al) optimized to maximize the internal quality factor (Q_i), while isolating interface-induced TLS losses from extraneous dissipation mechanisms. The circuit's purpose is to, then, carry out measurements of their respective S_{21} spectrums in order to extract Q_i at different temperatures. This calculation of $Q_i(T)$ is used in order to extract each material's characteristic loss tangent (δ^0_{TLS}).

Due to the scarcity of literature on the cryogenic microwave properties of diamond, 4H-SiC, and 6H-SiC, predicting the resulting Q_i prior to characterization presents a significant challenge. To ensure that at least one resonator per substrate operates within or near the critically coupled regime ($g \approx 1$), each chip layout incorporates a set of four $\lambda/4$ coplanar waveguide (CPW) resonators inductively coupled to a central CPW feedline, terminated by launchers on both ends. This configuration allows for the systematic exploration of external quality factors (Q_e) spanning distinct orders of magnitude.

The proposed circuit designs were modeled using the method-of-moments simulation software Sonnet Software. Simulations were leveraged to extract key electromagnetic parameters—specifically the effective permittivity (ϵ_{eff}), characteristic impedance (Z_0), and resonant frequency (f_r)—and to estimate Q_e directly from the simulated S_{21} scattering parameters. Following numerical optimization, the designed chips were fabricated via a lift-off process, optically inspected for physical defects, and evaluated within a dedicated cryogenic measurement framework designed to characterize dielectric TLS losses in the single-photon regime.

2 Theory background

In this chapter, key theoretical concepts needed to understand the general workflow, are introduced in order to understand the results provided later on. A section dedicated to general concepts of superconductivity will serve to understand the materials being used. Following this, a section describing the behavior of quarter-wave coplanar waveguide resonators will introduce and give a detailed explanation of the concepts and expressions used to make predictions for the overall design and analysis of the systems to be studied.

2.1 Superconductivity

This section will be dedicated to the explanation of specific concepts related to superconductivity relevant to this thesis. First, explaining basic phenomenology; then, touching on the subject of BCS theory and Cooper pairs; and finally, arriving at the concept of type-I and type-II superconductors.

2.1.1 Basic phenomenology

This section will follow the analysis shown by [3].

When talking about superconductivity, the first thought that comes to mind is conduction with a resistance exactly equal to zero; or rather, conduction with an infinite value of conductivity. And, even though this is, in reality, the most desirable key characteristic of superconducting materials due to the lack of resistive energy losses, the truth is that a completely pure metal with no impurities would also show this characteristic, as its temperature approached absolute zero, without being catalogued as a superconductor. The key differences between the two are as follows.

Firstly, a true superconductor experimentally shows a sudden drop in resistivity to exactly zero as shown in Fig. 1, at a given temperature called the critical temperature of a superconductor (T_c). This sudden drop, is an indicator of a clear phase transition between what is called a normal conductor and a superconductor.

Additionally, in the case of a superconductor, once the critical temperature is achieved, the magnetic field inside the material also becomes exactly zero; this phenomenon called

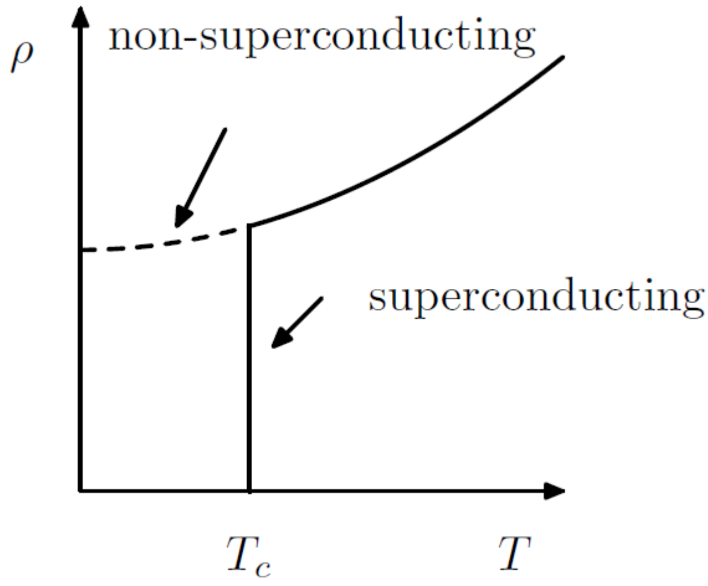

Figure 1: Phase transition diagram of a superconducting material. The resistivity is shown to drop violently at T_c (solid line), unlike a regular conductor (dashed line). Reproduced from [3].

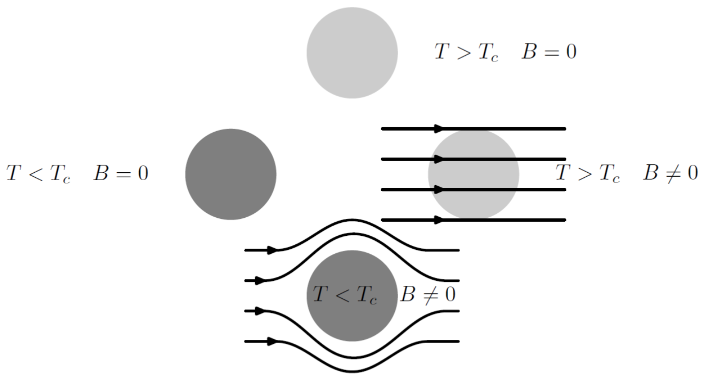

Figure 2: Representation of the Meissner Ochsenfeld effect. Independently of its initial state, the material will end up having an internal flux of exactly zero. Reproduced from [3].

the Meissner-Ochsenfeld effect consists in the superconducting material arriving at the same state of equilibrium with $B = 0$ inside it, independently of its previous state, as shown in Fig. 2. This means that whether there was an existing external magnetic field traversing the material or not, does not matter to its final state. It is worth noting that if the material did have an external magnetic field H going through its volume, in order to expel it, it would become magnetized the opposite way, such as $M = -H$; this is why superconductors can also be considered perfect diamagnets ($\chi = -1$).

Considering that the equipment used for resistivity measurements adds resistivity that proves to be difficult to exclude from the results, the Meissner effect is the preferred experimental way of proving that a certain material does, in fact, present superconductivity; since the measurement of magnetization proves to be less invasive to the material.

2.1.2 London penetration depth and superconductivity types

Of interest to this thesis is the classification of superconductors into two types, Type-I and Type-II, based on their response to external magnetic fields [3].

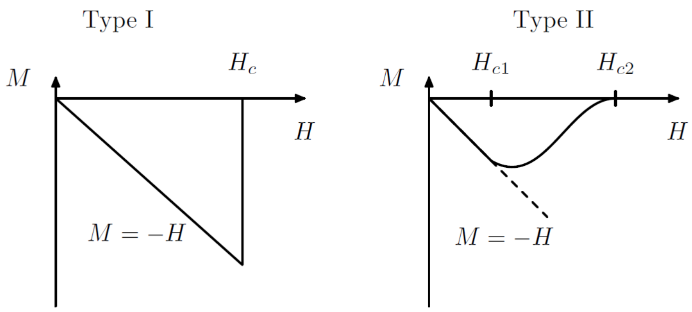

Figure 3: Visual representation of the abrupt transition of the internal flux, presented by a Type-I superconductor, in comparison with the smooth transition presented by a Type-II superconductor. Reproduced from [3].

In a Type-I superconductor, the same way there exists a critical temperature T_c or a critical electrical current I_c for which superconductivity is suddenly broken, there exists a critical magnetic field H_c for which resistivity ceases to be exactly zero and the Meissner effect ceases to occur. This means that the material's magnetization becomes zero as it is represented in Fig. 3, and therefore the lines of the external field H suddenly penetrate the material normally.

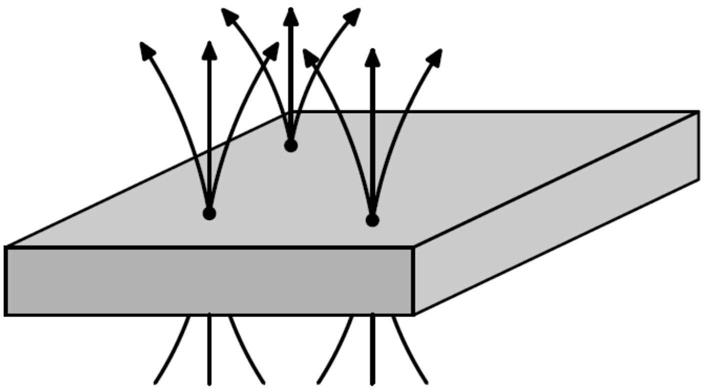

Figure 4: Schematics of the penetration of magnetic flux lines through a Type-II superconductor. Reproduced from [3].

On the other hand, on a Type-II superconductor this transition towards normal conductivity is much smoother. In this case, rather than taking one critical magnetic field H_c, two critical field values are studied, H_{c1} that represents the value of H for which M stops strictly behaving as $M = -H$; and H_{c2} that represents the value of H for which M exactly zero. As a consequence of this behavior, magnetic flux is able to partially penetrate the material in the interval between H_{c1} and H_{c2}, in the form of discrete flux vortices as it is represented in Fig. 4.

Around these vortices the magnetic field decays radially over the characteristic length known as the London penetration depth λ_L. In the context of this thesis, it is notable how this quantity might behave in thin-film superconductors.

According to [4], when a thin-film's thickness is smaller than the mean free path l[1], the mean free path behaves such as $l \approx t$ where t is the thickness of the metallic film. This is consequential, as the value of the effective λ in the dirty superconductor limit[2] (${\xi_0}^3 \gg l$) depends on l such as

$$\lambda = a\lambda_L\sqrt{\frac{\xi_0}{l}}. \tag{2.1}$$

Where a depends on the surface scattering type ($a \approx 1.33$ for diffusive scattering, and $a \approx 1.15$ for specular reflection [5]). This way, the effective penetration depth λ becomes bigger as the thickness of the thin-film is reduced. This way, [4] discusses how the thickness of the thin-film can be related to a transition in the superconductivity type of aluminum. In particular, they discuss how, as the thickness of the film becomes larger, in an interval of [54nm, 207nm], the apparition of flux vortices decreases up to a point where the material appears to become a Type-II superconductor. This phenomenon is also linked to other aspects of the geometry of the film; in this case, the geometry of the resonators, including the ratio between the thickness of the film and the width of the conductor of the CPW resonators used for measurements.

2.1.3 BCS theory and quasiparticles

In order to better understand the mechanisms of superconductivity, an understanding of the microscopic physics behind the behavior of these materials is needed. The Bardeen, Cooper and Schrieffer (BCS) theory, through the use of quantum many-body theory, does a good job at explaining and predicting the experimentally known phenomena surrounding superconductivity [3].

This framework treats the interactions between electrons inside the superconductor through the combined effect of screened Coulomb repulsion and an attractive, phonon-mediated interaction, together acting on the wave-function of the entire body of electrons. The screened Coulomb repulsion between electrons takes the form

[1]In the context of superconductors, the mean free path is defined as the characteristic length in between collisions between quasiparticle pair-constituent electrons and discontinuities such as phonons or impurities in the material [5].

[2]In general, evaporated Al thin films work in this regime, as the Al bulk coherence length is $\xi_0 = 1.6\,\mu\text{m}$.

[3]ξ is defined as the coherence length; which corresponds to the length after a collision where the order parameter is recovered [5]

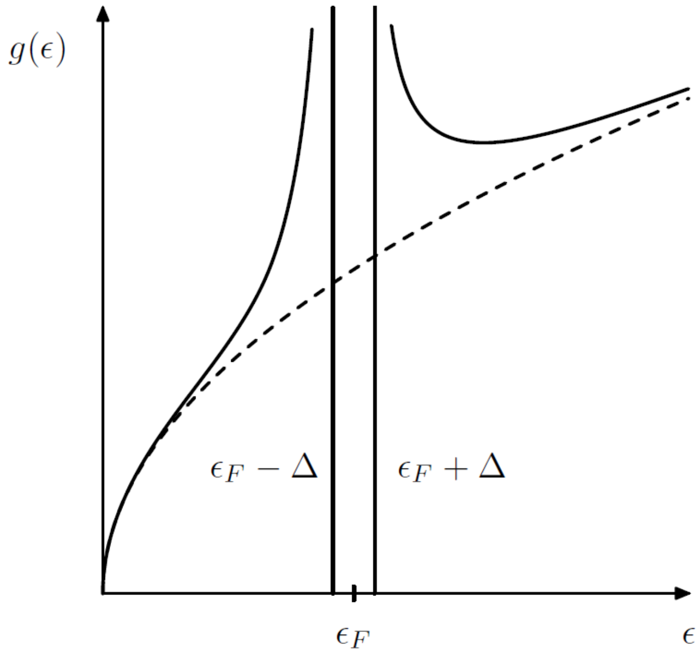

Figure 5: BCS density of states as a function of energy ϵ. This schematic shows the characteristic gap of 2Δ centered at the fermi level ϵ_F, where no quasiparticle states exist. Reproduced from [3].

$$V_{TF}(\mathbf{r}-\mathbf{r}') = \frac{e^2}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|} e^{-|\mathbf{r}-\mathbf{r}'|/r_{TF}}, \tag{2.2}$$

which is overcome, within a narrow energy window near the Fermi surface, by the attractive interaction mediated by lattice phonons. The net result is a weak effective attraction between electrons near the Fermi level, responsible for the formation of Cooper pairs; which are bound states formed by two electrons sharing a common wave-function.

This attractive interaction shows itself as a formation of an energy gap in the excitation spectrum,

$$2\Delta(0) = 3.52 k_B T_c, \tag{2.3}$$

which, as can be seen in Fig. 5, remains centered at the Fermi level. This result is especially notable, as it explains the robustness of the Cooper pair against dissipation.

Additionally, when one takes into account collisions with imperfections within the superconductor's volume or phonons, the results show a complex conductivity for the superconducting state [6]

$$\sigma = \sigma_1 + i\sigma_2. \tag{2.4}$$

Where the real part represents the energy losses due to collisions with phonons with energy of the order of the energy gap Eq. (2.3), as well as with imperfections in the material. These collisions can lead to Cooper pairs breaking, which produces two unpaired Bogoliubov quasiparticles. These quasiparticles behave closer to normal conductivity and therefore dissipate energy in an analogous manner to Ohmic losses.

2.2 Quarter-wave coplanar waveguide resonators

The following section will be dedicated to the theoretical explanation of the general behavior of quarter-wave coplanar waveguide resonators focusing on the mechanisms of resonance, resonance frequency, characteristic impedance, quality factor and energy loss mechanisms.

2.2.1 General CPW physics

In order to understand the physics behind quarter-wave CPW resonators, one must first understand what a microwave resonator is.

Usually, when the wavelength of the electromagnetic signal used is much bigger than the size of the elements used to build a circuit (which is the case for domestic power transmission at around 50Hz, which translates to wavelengths of around 6000km), the lumped element model is used for the analysis of electric circuits. When working in this regime, it is possible to build resonators such as the RLC resonator. This system consists in a circuit containing a resistor, R, an inductor, L and a capacitor C. Using these elements in different arrangements, it is possible to arrive to several different devices, the most emblematic of which, are the series and the parallel RLC resonators.

As their name suggest, these circuits are characterized by the fact that the electric field generated by the capacitor and the magnetic field generated by the inductor both oscillate the same way a mechanical oscillator's kinetic and potential energy would oscillate. In this kind of oscillator, resonance is found at the frequency where the input impedance (which is defined as the complex ratio between voltage and alternating current) is purely real. This frequency shows to be such as

$$\omega_0 = \frac{1}{\sqrt{LC}}. \tag{2.5}$$

Similarly, when the size of the elements studied is comparable to the wavelength of the electromagnetic field (which is the case for the microwave spectrum, corresponding to wavelengths on the order of centimeters), the distributed-element model is used. This model describes a transmission line in terms of per-unit-length parameters resistance (R'), inductance (L'), conductance (G'), and capacitance (C') assigned to infinitesimal

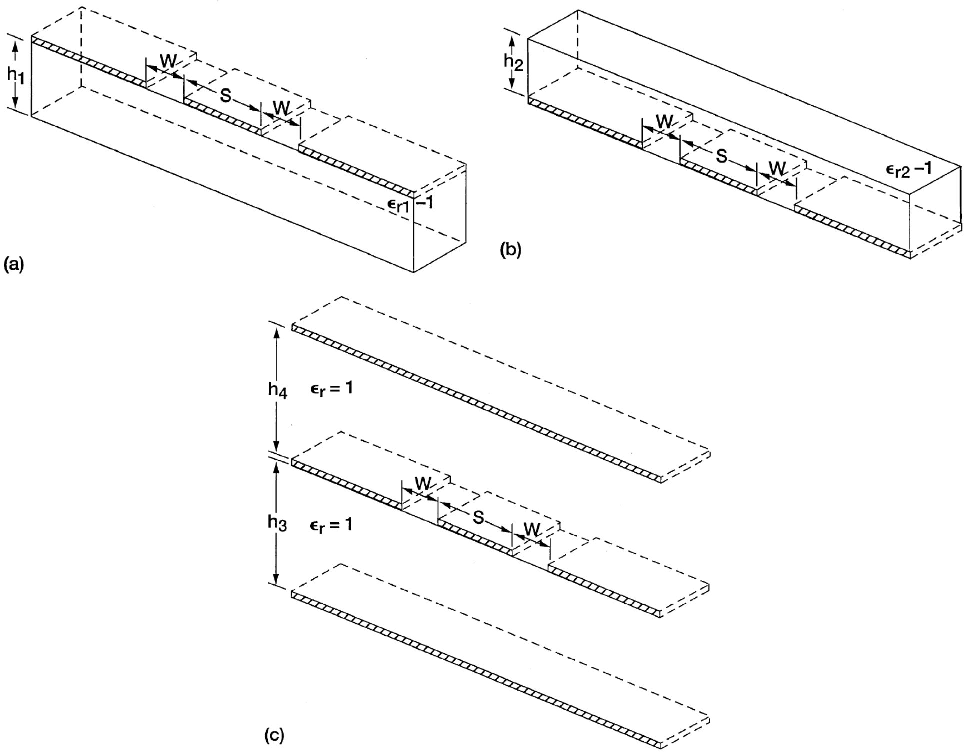

Figure 6: Schematics of the three different partial capacitances calculated using conformal mapping techniques for a coplanar waveguide sandwiched between two dielectric volumes. Reproduced from [7].

segments of the line, analogous to the lumped-element model. Taking the continuum limit of these segments yields the telegrapher's equations, which govern voltage and current as functions of position along the line. This model takes into account how the voltage and subsequent current change along the surface of each element; revealing aspects of the circuit electrodynamics that are lost in lumped element circuitry.

In the case of the distributed element, the resonators most represented in the literature are resonant cavities, $\lambda/2$ CPW resonators and $\lambda/4$ CPW resonators. For the sake of this study, only the case of quarter-wave coplanar waveguide resonators will be explored.

For distributed-element structures with simple field symmetry, such as coaxial lines or parallel-plate capacitors, the per-unit-length capacitance, and hence ϵ_{eff} and Z_0, can be obtained directly from Gauss's law in closed form. Coplanar waveguides, however, lack this symmetry. This is because the electric field is distributed inhomogeneously between the substrate and the air above the slots, and no simple closed surface captures the geometry. Conformal mapping techniques, are therefore used to obtain closed-form expressions for C' by mapping the CPW cross-section onto an equivalent geometry where the capacitance is tractable.

In this particular case, conformal mapping proves useful as the equations to be described are the Laplace equations, which are notoriously invariant to conformal transformations.

According to the results shown by [8] and [7], by using the Christoffel-Schwartz transformation in order to compute the total capacitance as the sum of the capacitances computed in the partial regions shown in Fig. 6 such as:

$$C_{CPW} = C_1 + C_2 + C_{air}. \tag{2.6}$$

And using the following relationship [8]:

$$\epsilon_{eff} = \frac{C_{CPW}}{C_{air}}. \tag{2.7}$$

One can arrive to the following results for the effective permittivity and the characteristic impedance when taking into account a coplanar waveguide with a grounded finite substrate [9]:

$$\epsilon_{eff} = \frac{1 + \epsilon_r \frac{K(k_0')}{K(k_0)} \frac{K(k_1)}{K(k_1')}}{1 + \frac{K(k_0')}{K(k_0)} \frac{K(k_1)}{K(k_1')}} \tag{2.8}$$

and

$$Z_0 = \frac{60\pi}{\sqrt{\epsilon_{eff}}} \frac{1}{\frac{K(k_0)}{K(k_0')} + \frac{K(k_1)}{K(k_1')}} \tag{2.9}$$

Where $K(k_0)$, $K(k_0')$, $K(k_1)$ and $K(k_1')$ are complete elliptic integrals, whose modulus come given by:

$$k_0 = a/b \tag{2.10a}$$

$$k_0' = \sqrt{1 - k_0^2} \tag{2.10b}$$

$$k_1 = \frac{\tanh(\frac{\pi a}{4h})}{\tanh(\frac{\pi b}{4h}} \tag{2.10c}$$

$$k_1' = \sqrt{1 - k_1^2} \tag{2.10d}$$

As a result of this, the phase velocity of light through the circuit will as

$$v_{ph} = \frac{c}{\sqrt{\epsilon_{eff}}}. \tag{2.11}$$

Finally, if one considers a transmission line with an open end and a shorted end; which is equivalent to having boundary conditions of $Z(0) = 0$ and $Z(l) \to \infty$; one arrives to the following condition [10]

$$l = (2n+1)\frac{\lambda}{4},\ n \in \mathbb{N}. \tag{2.12}$$

Hence, the length of the resonator would be

$$l_{total} = \frac{v_{ph}}{4 \cdot f_r}. \tag{2.13}$$

It is also worth noting the importance of properly taking into account the impedance in a resonator's design. According to [11, Chapter 5], the reflection coefficient near resonance of a quarter-wave CPW resonator is described as

$$|\Gamma| \approx \frac{|Z_L - Z_0|}{2\sqrt{Z_0 Z_L}} |\cos\theta|. \tag{2.14}$$

Where Z_0 is the characteristic impedance of the transmission line, as seen in Eq. (2.9), and Z_L is the loading impedance, which is defined as the impedance value terminating the transmission line; which in our case matches the impedance of the coaxial cable feeding the signal. Analyzing the expression, one can notice that the closer the characteristic impedance is to the load impedance, the lower the reflection coefficient will be, which is desirable when working with transmission readouts.

One way to study the resonance frequency and overall physics of a coplanar waveguide resonator circuit are scattering parameters, often represented through the scattering matrix such as [11]

$$\begin{pmatrix} V_1^- \\ V_2^- \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} V_1^+ \\ V_2^+ \end{pmatrix}.$$

In this notation, the subindices i and j stand for the input and the output port, and the superindices + and - refer to the direction from which a wave is arriving to a port (In this case + means the wave is going away from the port and - means it is coming into the port). The elements of the matrix can then be interpreted as reflection terms when $i = j$ and transmission terms when $i \neq j$.

In the case of a quarter wave resonator, the main object of study is the behavior of the S_{21} coefficient. This element of the scattering matrix represents the transmission of signal from the input port into the output port. In the case of the systems at hand, when resonance is achieved, a significant part of the energy going through the feedline gets transferred to the resonator, and thus leading to a dip in the S_{21} coefficient. Exploring this behavior can prove rather useful to study characteristics of the circuit or the materials that conform it, such as capacitances, inductances, and energy losses.

2.2.2 Quality factor and energy loss

Energy losses are mainly quantified using the concept of quality factor (Q). This concept is directly tied to the overall performance of a superconducting circuit. The loaded quality factor of a given circuit can be separated into several quality factors corresponding to the several loss mechanisms that can be found. This relationship can be expressed as [1]

$$Q_L^{-1} = Q_{TLS}^{-1} + Q_c^{-1} + Q_{qp}^{-1} + Q_{vortex}^{-1} + Q_{rad}^{-1} + \dots \tag{2.15}$$

Even though there is a great quantity of energy loss channels, for the sake of this study only the ones shown will be explained; going into greater detail for Q_{TLS}, Q_{vortex} and Q_c.

In this case, Q_{qp} stands for the energy losses derived from non-equilibrium quasiparticles in the form of unpaired electrons formed inside the superconductor that experience local resistivity when moved due to the AC field going through the circuit. These quasiparticles can form due to a variety of reasons; extending from thermal radiation, to high energy photons coming from cosmic radiation; and is one of the main interests of study in the field of energy losses related to superconducting quantum computing.

Furthermore, Q_{rad} refers to the energy losses caused by the radiation of the electromagnetic field into free space or into the chip package.

Likewise, Q_{vortex} is derived from the contribution of the movement of vortices formed in a type-II superconductor.

More importantly to the subject of this study is the contribution of two-level-systems (TLS). Q_{TLS} represents the energy loss contributions of the different TLS that appear along the interfaces between materials. Such interfaces are the air-substrate interface, the air-metal interface and the substrate-metal interface. The latter is usually the one with the highest contribution. This quantity can be described by

$$Q_{TLS} = \sum_i p_i \delta_i. \tag{2.16}$$

Where p_i represents the participation ratio of the respective interface; and δ_i corresponds to the loss tangent, or the weight of the energy loss, corresponding to that same interface. It is relevant that the value of δ_i is not a fixed characteristic of the materials. Especially, the value of δ_{TLS}^0 is also sensitive to temperature as follows [10]

$$\delta_{TLS} = \delta_{TLS}^0 \tanh\left(\frac{\hbar\omega}{2k_B T}\right). \tag{2.17}$$

This relationship represents how the absorption of the energy of the resonators is directly linked to the occupation statistics of the TLS states, which follow the Boltzman occupation numbers for a TLS.

Furthermore, when working in the low-photon-number regime this quantity stays at the intrinsic TLS loss tangent value δ_i^0. On the other hand, at high drive powers, the two level systems become saturated at a much faster rate than their respective relaxation times; effectively decreasing the losses caused by this channel. This effect can be seen described by [10]

$$\delta_i(\langle n \rangle) = \frac{\delta_i^0}{\sqrt{1 + (\langle n \rangle / n_c)}}. \tag{2.18}$$

Where $\langle n \rangle$ represents the number of photons inside the resonator and n_c represents the critical photon number needed to saturate the TLS.

All of these contributors to the loss of energy in the system are part of what is called the internal quality factor. These are losses that are intrinsic to the circuit materials, environment and geometric configuration.

Separately, the loaded quality factor also takes into account what is called an external quality factor (Q_e) or coupling quality factor (Q_c). This value quantifies the rate of energy loss caused by the coupling of the resonator to the feedline for readout. As it is discussed in [11], when working with CPW resonators, this coupling can be done in two different ways depending on the dominating electromagnetic quantity at the end of the coupling point.

If the resonator terminates in a shorted end near the coupling point, then there would be a current antinode and a voltage node at the end of the line (maximum current and minimum voltage). Hence, the coupling is given by the mutual inductance of the current flowing through the resonator and the current of the feedline; this is what is called inductive coupling.

On the other hand, If the resonator terminates in an open end near the coupling point, then there would be a voltage antinode and a current node at the end of the line (maximum voltage and minimum current). As a result, the coupling arises from the capacitance between the resonator and the feedline; this is what is called capacitive coupling.

In both scenarios, coupling strength is governed by the distance separating the end nearest end of the resonator to the feedline and by the length of the segment of the resonator at coupling distance to the feedline. The closer the resonator or the longer the coupling segment, the stronger the energy exchange will be. Therefore, the coupling will be stronger, and the energy loss rate will increase; which will lead to a decrease in the value of Q_c.

As a result of the difference in the origin of energy losses between Q_c being an external loss factor, and the other contributors to the quality factor mentioned being internal loss factors; it is useful to make a difference between them as internal quality factor (Q_i) and external quality factor (Q_e) such as

$$Q_L^{-1} = Q_i^{-1} + Q_e^{-1}. \tag{2.19}$$

This classification is especially useful for experimental setups, as the coupling coefficient ($g = Q_i/Q_e$) becomes relevant to the quality of the signal. This ratio represents the relationship between the strength of the coupling between the resonator and the feedline, and the energy losses intrinsic to the circuit. Through this quantity one can consider three different working regimes.

Firstly, if the strength of the coupling is too strong, that is, the losses due to coupling are much greater than the intrinsic losses of the circuit ($g \gg 1$), the system is said to be overcoupled.

On the contrary, if the strength of the coupling is too weak, that is, the losses due to coupling are much smaller than those intrinsic to the circuit ($g \ll 1$), the system is said to be undercoupled.

Finally, if the losses due to the coupling are comparable to the losses intrinsic to the circuit ($g \approx 1$), the system is said to be critically coupled.

The importance of this ratio becomes evident when considering the S_{21} frequency spectrum for a hanger style topology like the one used in this thesis; described by [10] as follows

$$S_{21}(f) = 1 - \frac{(Q_l/Q_c)}{1 + 2iQ_l/Q_c(f/f_r - 1)}. \tag{2.20}$$

This expression can be expanded such as

$$S_{21}(f) = 1 - \frac{g}{1+g}\left[\frac{1}{1+\left(2Q_L\frac{f-f_r}{f_r}\right)^2} - \frac{2iQ_L\frac{f-f_r}{f_r}}{1+\left(2Q_L\frac{f-f_r}{f_r}\right)^2}\right]. \tag{2.21}$$

This expression reveals several important features of the resonator response. Firstly, it is notable that when the frequency is set to the resonant frequency the expression is reduced to

$$|S_{21}(f_r)| = 1 - \frac{g}{1+g}. \tag{2.22}$$

From this expression one can see how there are three possible limits:

- **Undercoupled ($g \ll 1$):** the value of S_{21} approaches 1; meaning that none of the signal is transferred to the resonator.
- **Critically coupled ($g \approx 1$):** $g \approx 1 \implies S_{21} \approx 1/2$; meaning that half of the signal is transmitted and half of the signal is retained by the resonator.
- **Overcoupled ($g \gg 1$):** the value of S_{21} approaches 0; meaning that the entirety of the signal is transferred to the resonator and none is transmitted.

In order to further understand the S_{21} magnitude and phase spectrum, one must study the full expression Eq. (2.21). At first glance, it is notable that, at resonance, the imaginary term disappears, while the real term reaches its maximum value, which can be observed in Eq. (2.22). It proves especially useful to study the S_{21} spectrum as a complex number of the form $S_{21} = |S_{21}|e^{i\phi}$; where the modulus is the quantity represented in the magnitude spectrum of the S_{21} response, and ϕ is a phase that arises from a frequency dependent phase response introduced by the resonator into the transmitted signal.

From this it is evident that the value of Q_L has a significant effect on the width of the S_{21} magnitude dip. This means that as Q_L becomes bigger the dip becomes narrower, and in contrast, as it becomes smaller, the dip becomes broader. Combined with the discussed coupling regimes, it is of interest to this thesis to study how the S_{21} magnitude response will respond with changes in Q_c.

As we have seen, increases in g lead to a deeper dip in S_{21} which makes the signal easier to measure. Additionally, increases in Q_L lead to a narrower signal, which makes the analysis of the signal easier and more exact. This is especially important as g is inversely related to Q_c; the opposite of Q_L which increases with Q_c as seen in Eq. (2.15). Hence, different coupling regimes are more desirable for different uses. In the case of sensing, the dip depth is more relevant than its width, and therefore an overcoupled resonator is desirable. On the other hand, for characterization or device control the dip width and data analysis more relevant than the sensitivity for the resonator; as a result critically coupled resonators are the most desirable, since undercoupled resonators usually can not be measured with standard equipment, and critical coupling maximizes the depth of the dip in the S_{21} magnitude response while leaving its shape easy to analyze.

Finally, using these results which facilitate the calculation of Q_i, one can use the loss tangent's dependency on temperature ($\delta_{TLS}(T)$) as a way to extract a material's δ^0_{TLS}. In order to do so, one can use the following relationship [10]:

$$\frac{1}{Q_i(T)} = F\delta^0_{TLS} \tanh\left(\frac{hf_r}{2k_B T}\right). \tag{2.23}$$

In this case, F is the filling factor, which represents the ratio of electromagnetic field stored in the TLS-host material to the total electric energy stored by the resonator. This way, carrying out a temperature sweep of the value of Q_i one can then fit the expression to find $F\delta^0_{TLS}$. According to [10], the value of δ^0_{TLS} may be extracted by calculating F through EM simulations or contour, or volume integrals of the electric field over the host material region.

3 Resonator design and simulations

The following chapter will focus on the explanation of the design and simulation process followed to arrive to the final chip design desired for the execution of the proposed experiment. The first section will be dedicated to the explanation of the general layout as well as the role of each parameter taken into account; then, a second section will focus on the final parameters chosen based on the simulations carried out in order to have a better estimation of the resonator's characteristics.

3.1 Layout design

The first step for the study of the substrates at hand was the design of a fabrication mask dedicated to each substrate. For the design process, many parameters and constraints were considered. Firstly, constraints given by the laboratory equipment were taken into account. These constraints came down to the characteristic impedance of the coaxial cables that would later on feed and measure the signal going through the chips, which was a standard value of 50Ω; the frequency range that could be measured by the setup, which goes between 4 GHz and 8 GHz, because of the presence of filters and circulators in the dilution fridge; and the xy-plane precision estimated for the fabrication process, which was of 1 µm.

The chip designs were thought for the respective sizes of each of the substrates available for this research. In the case of both SiC substrates, each chip has dimensions of 10.00 × 10.00 mm and 0.35 mm of thickness; and, in the case of the diamond substrate, the chip had dimensions of 4.5 × 4.5 mm and 0.5 mm of thickness.

Once all of these factors were properly taken into account, it was decided that a set of four $\lambda/4$ CPW resonators inductively coupled to the feedline, in the range between 6 GHz and 8 GHz, separated by at least 200 MHz would be best to both optimize the measurement region and avoid crosstalk between resonators.

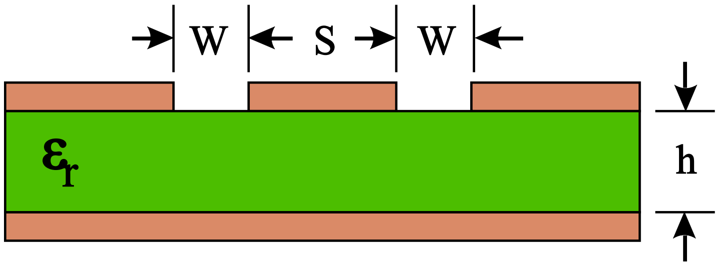

Figure 7: Physical parameters required for calculations using conformal mapping in Eqs. (2.8) and (2.9). Reproduced from [12].

Physically speaking, the main parameters taken into account were the center conductor width and the gap to the ground of the waveguide such as what is shown in Fig. 7 which, as discussed in Section 2.2.1, are the main contributing factors to the resonator's Z_0; as well as the total length of the resonator and the distance between the resonators and the feedline, which would in turn be the main contributing factors to the resonance frequency and the Q_e accordingly.

A first estimation of the parameters was done using the expressions Eqs. (2.8) and (2.9) for the ϵ_{eff} which is directly related to the f_r and the Z_0. At the same time, an initial set of reference parameters was taken from a study carried out by [13], which proved to be the layout with the highest Q_i in the aforementioned review by [14], in the particular case of 2D aluminum circuits. This initial set of parameters was of 20 µm and 10 µm for the center conductor width (w) and the gap to ground (g) respectively, and an aluminum thickness of 300nm. Due to the reference setup being of aluminum over silicon, the values for w and g were adapted for each of the substrates to be studied.

As it is shown Eqs. (2.8) and (2.9), both the ϵ_{eff} and the Z_0 come mainly given by the geometry of the CPW and the ϵ_r of the substrate. Therefore, the parameters mentioned in the previous paragraph were adapted for the values of ϵ_r for diamond, 4H-SiC and 6H-SiC that were retrieved from references [15, 16, 17] respectively, which was 5.7 for diamond; and in the case of the two kinds of SiC, due to being anisotropic materials, 9.77 for the ordinary axis and 10.2 for the extraordinary axis for 4H-SiC; and for 6H-SiC 9.66 for the ordinary axis and 10.3 for the extraordinary axis; in the cases of SiC, the xy-plane of the substrate was taken as parallel to the ordinary axis for the purpose of calculations. The adjusted values of both w and g were finally 20 µm and 10 µm respectively for both SiC orientations, and 20 µm and 4 µm respectively for diamond.

3.2 Simulation iterations

After estimating these initial parameters, a series of numerical simulations were carried out using the method of moments software Sonnet Software. These simulations served

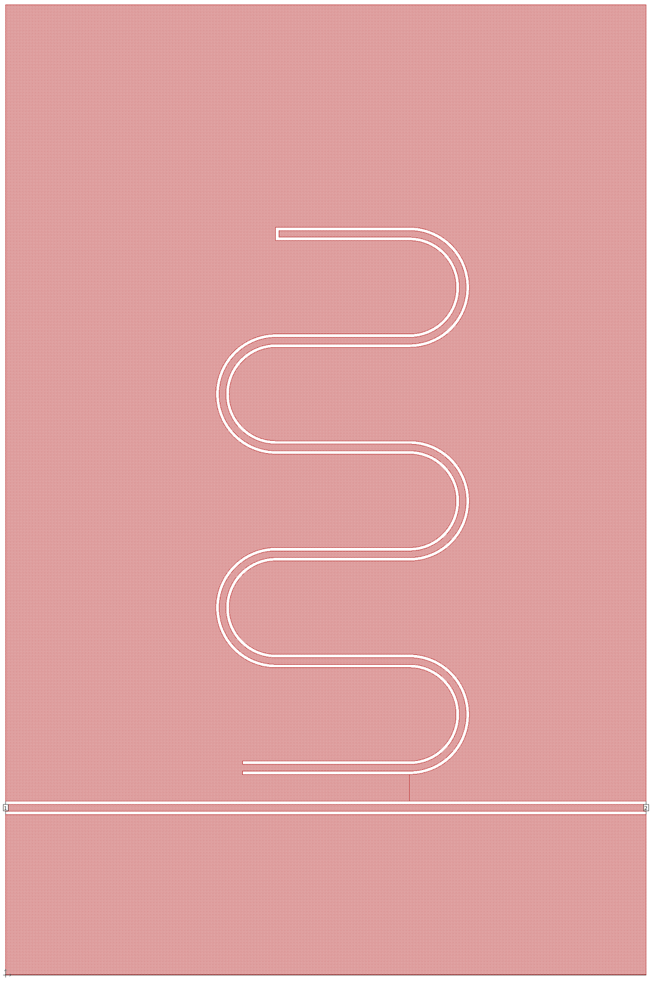

Figure 8: Reduced layout used for Sonnet simulations. The reduced ground surface serves to reduce simulation time without loss of physical accuracy.

as a way to tweak the estimation of the f_r of each resonator; as well as a way to study the behavior of the relationship between Q_e and the coupling distance (d_c). In order to carry out these simulations a reduced layout containing a single resonator and the feedline measuring 1920 mm $\times$ 2907.5 mm was used, as seen in 8. This layout was divided using a rectangular mesh with a cell size of 2.5 μm $\times$ 2.5 μm; taking into account that the smallest distance in the SiC designs was 10 μm; and in the case of the diamond design 4 μm. It is of interest that even though a conformal mesh would have provided more accurate results, a rectangular mesh was used due to great increases of simulation time with the former. In order to carry out the desired assessment of the chips including a general study of the S21 response, the Z_0 and the ϵ_{eff}, adaptive frequency sweeps were implemented in an interval close to the predicted f_r.

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This process was done in an iterative manner; where, for each resonator of each substrate, firstly a coarse frequency sweep of 300 points was done in order to adjust the length of the resonator taking into account the ϵ_{eff} calculated by Sonnet and get closer to the desired f_r. Then, with the updated layout, a new coarse simulation was done in a 200MHz interval around an educated estimation of the f_r in order to get a first indication of the power dip in the signal. Then, finer simulations of 600 and 800 points were done in progressively smaller intervals around the f_r until the dip stopped growing vertically, which meant that the signal had been finely resolved closely around the f_r. Finally, the resolved signal was analyzed using the method developed by [4], which was useful to have a proper estimation of Q_e.

Through this process, the final parameters for the resonators were achieved. They can be found in Table 1.

3.3 Simulation results

Table 1: Final parameters achieved for resonators with 4H-SiC, 6H-SiC, and Diamond substrates. All parameters were iteratively simulated in order to arrive to the desired conditions.

	w (µm)	g (µm)	l_{total} (µm)	d_c (µm)
4H-SiC	20	10	{5005, 4755, 4529, 4323}	{80, 70, 70, 25}
6H-SiC	20	10	{4973, 4725, 4500, 4295}	{80, 70, 70, 25}
Diamond	20	4	{5988, 5682, 5411, 5165}	{50, 40, 40, 20}

After the simulations were carried out, the overall behavior of the resonator described in Section 2.2.1 was studied. This comprehensive study included the general electromagnetic behavior of the layout, such as voltage and current density along the CPW, and the resulting values of ϵ_{eff}, and Z_0, which were consequently compared to those calculated from [9] used for the initial estimations mentioned in Section 3.1. Furthermore, it included an in-depth analysis of the S_{21} spectrum following the method exposed in [4], from which an estimation of f_r and Q_c was gotten; the corresponding fits are provided in Figs. 15 to 17. The numerical results of the simulations can be seen in Tables 3 to 5.

Table 2: Values of Z_0 and ϵ_{eff} for resonators with 4H-SiC, 6H-SiC, and Diamond substrates. All values where extracted from Sonnet simulations.

	4H-SiC Th	4H-SiC Sim	6H-SiC Th	6H-SiC Sim	Diamond Th	Diamond Sim
$Z_0(\Omega)$	52.18	55.23	51.91	55.25	51.01	55.51
ϵ_{eff}	5.33	5.63	5.388	5.63	3.35	3.52

Table 3: Data extracted from fitting the simulated S_{21} [4] spectrum for resonators with 4H-SiC substrate. The results are compared to those gotten from analytical expressions.

Resonator length (µm)	$f_{r,theory}$ (GHz)	$f_{r,simulation}$	Coupling distance (µm)	$Q_{c,simulation}$
5005	6.49	6.22	80	1.5×10^6
4755	6.83	6.54	70	4.64×10^5
4529	7.17	6.86	70	4.2×10^5
4323	7.51	7.19	25	11.75×10^3

Table 4: Data extracted from fitting the simulated S_{21} [4] spectrum for resonators with 6H-SiC substrate. The results are compared to those gotten from analytical expressions.

Resonator length (µm)	$f_{r,theory}$ (GHz)	$f_{r,simulation}$	Coupling distance (µm)	$Q_{c,simulation}$
4973	6.50	6.27	80	1.5×10^6
4725	6.83	6.58	70	4.56×10^5
4500	7.18	6.86	70	4.2×10^5
4295	7.52	7.23	25	11.6×10^3

It is interesting to point out how the simulated values of ϵ_{eff} and Z_0 systematically land higher than the ones calculated through the expressions derived from [7]. This can be explained by the fact that in Wadell's equations[9], there are several approximations that take place. Firstly, the ground surface around the resonator is approximated to be infinite; and, secondly, the metal is approximated to have a null thickness.

These differences between the expected results and the ones retrieved from the simulations were studied. In the case of the metal thickness approximation, in [7], it is shown how, if the thickness of the metal is considered, as the metal thickness goes up, the predicted value of ϵ_{eff} becomes higher than the one expected from the initial conformal mapping expressions, and the value of Z_0 becomes lower than the one expected. On the other hand,

Table 5: Data extracted from fitting the simulated S_{21} [4] spectrum for resonators with diamond substrate. The results are compared to those gotten from analytical expressions.

Resonator length (μm)	$f_{r,theory}$ (GHz)	$f_{r,simulation}$	Coupling distance (μm)	$Q_{c,simulation}$
5988	6.84	6.59	50	9.18×10^5
5682	7.21	6.94	40	2.41×10^5
5411	7.57	7.30	20	24.7×10^3
5165	7.93	7.64	40	2.01×10^5

for the ground surface size approximation, it was found that, according to [14], as the ground surface around the conductor line is accounted for as finite, the value of Z_0 goes up the smaller the ground surface size; this approximation did not seem to have a noticeable effect on the value of ϵ_{eff}. This last approximation was tested through a test run using a layout with a bigger ground size, closer to the real size of the chips. The results adjusted to the value of Z_0 decreasing with the larger ground. In the same way, test simulations were run that proved the value of ϵ_{eff} to increase as the thickness becomes bigger.

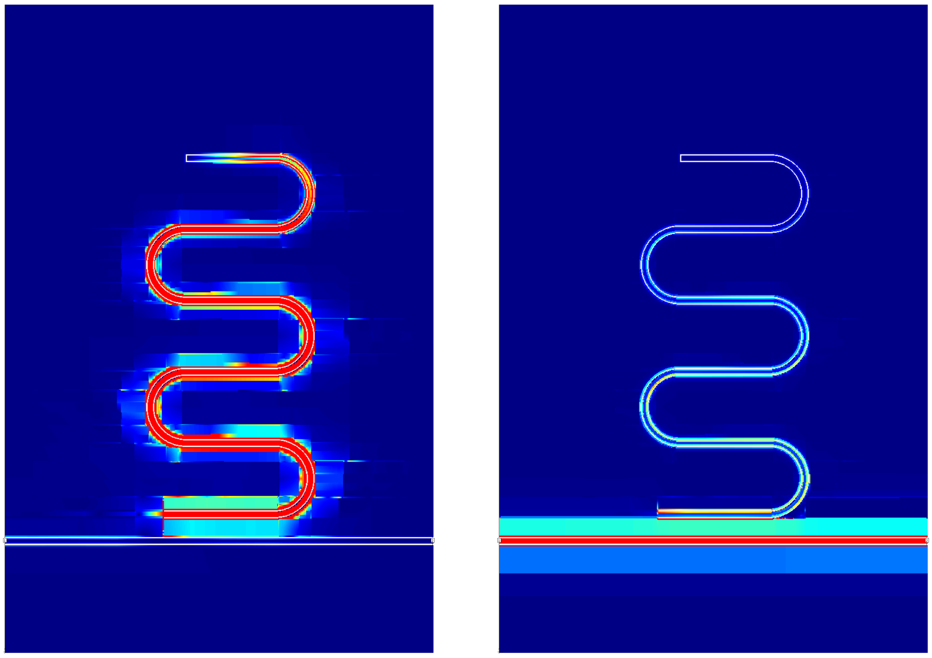
Being that the value of Z_0 was higher than the one predicted from the initial expressions it is thought that the effects of the 300nm thickness were negligible in comparison to the effects of the increase in ground size. This can be mainly explained by the fact that the size of the ground in the layouts used for the simulations was around a fifth of the real size in the x-axis and a third of the real size in the y-axis, and a thickness of 300nm results in a very small relation between the cross-section surface and the total length of the resonator.

Furthermore, the current density of the circuit can be seen in Figs. 9a, 9b and 18a to 18d. These results become useful for confirmation of several points.

First, these results confirm that the resonator is only hosting the fundamental mode, as higher-order modes would be undesirable.

Second, they confirm the boundary conditions discussed in Section 2.2.1. At the shorted end, the impedance approaches zero, forcing the voltage to zero ($Z(0) \approx 0 \implies V(0) = 0$); which, by continuity of power flow, implies that the current there must instead be maximal. At the open end, the impedance diverges, forcing the current to zero ($Z(l) \to \infty \implies I(l) = 0$), while the voltage, and correspondingly the electric field, reaches its maximum. This is precisely the current maximum at the short and current minimum at the open end, visible in the 4H-SiC case shown in Fig. 9.

Finally, this agreement extends to the three substrates, which was expected due to the analytical behavior only being explained through the geometry of the system and the only effects caused by the ϵ_r being on the f_r and subsequently on the total length of the resonator rather than the shape of the standing wave.



(a) 4H-SiC, in resonance **(b)** 4H-SiC, out of resonance

Figure 9: Simulated surface current distribution in resonance (top row) and out of resonance (bottom row) for a representative resonator on each substrate. The distribution is dominated by the CPW geometry and is essentially substrate-independent. As discussed in Section 2.2.1, out of resonance, there is no energy transmission from the feedline to the resonator, and therefore no current density inside it. Also, in resonance, there is a maximum current density at the shorted end nearest to the feedline and a current density minimum at the open end, where there is a voltage maximum.

3.4 Fabrication of the resonators

3.4.1 Fabrication runs

Once the final parameters were agreed to fit the general interests of this thesis they were sent out to the Max Planck Institute for Physics (MPP), in Munich, Germany, which supplied the substrates to be studied. These substrates had the following specifications:

- **Diamond substrate:**

  Quantity: 2

  Dimensions: $\sim$ (4.5 mm $\times$ 4.5 mm $\times$ 0.50 mm)

  Fabrication: Chemical vapor deposition (CVD)

- **4H-SiC substrate:**

  Quantity: 2

  Dimensions: $\sim$ (10.00 mm $\times$ 10.00 mm $\times$ 0.35 mm)

Fabrication: Chemical vapor deposition (CVD)

- **6H-SiC substrate:**

  Quantity: 2

  Dimensions: $\sim$ (10.00 mm $\times$ 10.00 mm $\times$ 0.35 mm)

  Fabrication: Chemical vapor deposition (CVD)

The fabrication process carried out at MPP was divided in two runs. Firstly, an attempt using wet etching over a positive resist was made. This attempt turned out to be inconsistent due to the viscosity of the etchant being too high for the 10 μm and 4 μm crevices and therefore unreliable. The substrates then a cleaning process in order to undergo a second fabrication run.

Secondly, a lift-off run was tried. This second run resulted in a successful fabrication attempt for the SiC substrates, where the shape of the resonators was properly resolved, and the resist could be properly developed out of the sample's surface. For the diamond substrates both runs presented complications caused by the smaller dimensions of the gaps ($g = 4\,\mu\text{m}$).

The process of the first run was carried out with the following steps:

1. The substrates were cleaned using an HF bath, which is generally crucial to ensure a clean fabrication surface.
2. An even 300 nm thick Al film was deposited on the samples using electron-beam evaporation under high vacuum conditions (UHV e-beam evaporation).
3. Spin-coating using micro resist technology's positive resist ma-P 1215 was done on each substrate.
4. The resist was exposed using a Heidelberg Instruments' μMLA laser writer following the layout shown in Fig. 10. This way the area where the Al coating was desired was protected by the resist.
5. The resist was developed using micro resist technology's ma-D 532/S.
6. Wet etching was carried out using phosphoric-acetic-nitric acid mixture (PAN etching).
7. The resist was stripped

Furthermore, the process of the second run went as follows:

1. The substrates were cleaned using an HF bath.

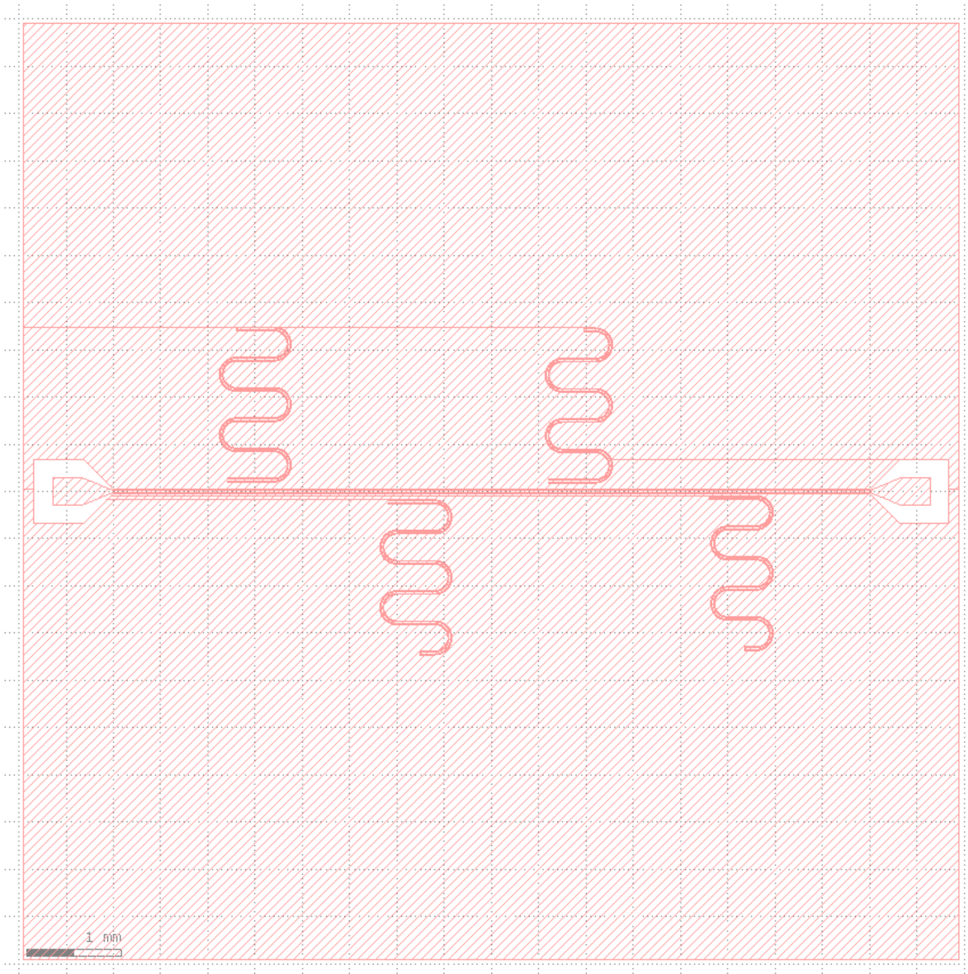


Figure 10: 4H-SiC Positive layout used for the wet etching process. The layout includes four $\lambda/4$ CPW resonators, inductively coupled to the central feedline. Launchers are added at each end of the feedline to facilitate wire-bonding. Each resonator can be seen to have distinct total lengths and distancing from the feedline. The colored areas protect the Al film, while the white gaps let the etchant through to generate the pattern.

2. Spin-coating using micro resist technology's negative resist ma-N 1440 was done on each substrate.

3. The resist was exposed using the aforementioned µMLA laser writer following the layout shown in Fig. 11. This way, the area of the substrate where the aluminum film was desired was left exposed, and the metal would not stick to the areas where the resist remained.

4. The resist was developed using micro resist technology's ma-D 333/S.

5. An even 300 nm Al film was deposited using the same UHV e-beam evaporation technique used in the first run.

6. The remaining resist was lifted off along with the Al deposited on top of it.

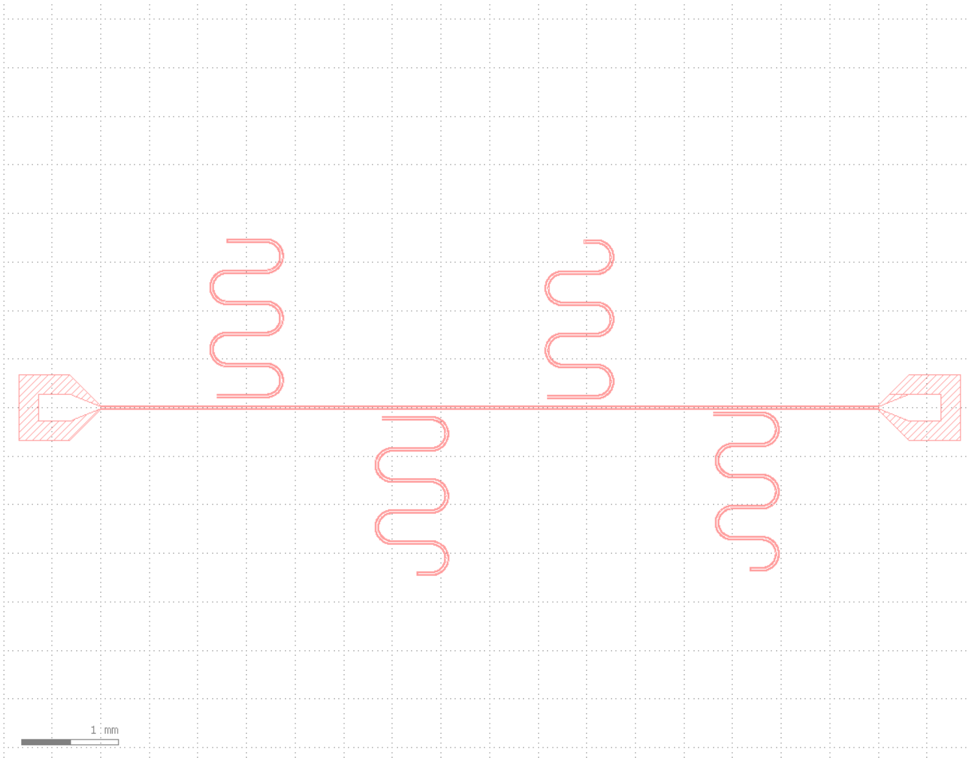


Figure 11: 4H-SiC Negative layout used for the lift-off process. This layout is the complementary (inverted) pattern to the positive mask. The colored regions of the resist become insoluble as they are exposed to the laser, protecting those areas from Al evaporation. Then the resist is removed leaving the pattern behind.

3.4.2 Fabrication results

Once the second fabrication run was finished, the four SiC chips were checked with a microscope at MPP, where they looked to be ready for measurements, and were sent to IFAE's QCT workshop. Once they arrived they were revised using an optical microscope in order to have an idea of the chip's viability for measurement. In particular, the chips were checked for any short-circuit or discontinuities caused by a broken CPW line or by faulty fabrication; as well as debris or pieces of resist remaining due to an incomplete strip.

Visual inspection under the microscope revealed defects in three of the four chips. Only one 6H-SiC chip was found to be in optimal conditions and was selected for measurement.

Due to fabrication and clean room schedule constraints, images of the chips are not available for discussion in this work.

3.5 Measurements planning and setup

Once the chips were obtained and available for measurement, it was of great importance to adjust the measurement setup to the experiment at hand. For this, the key aspects of the experiment had to be taken into account. These included the fact that the experiment was thought for superconducting Al, which meant its critical temperature (T_c) had to be taken into account; and the kind of measurements to be carried out, which were frequency and power sweeps seeking an in-depth study of the S_{21} spectrum of the resonators in order to then extract the internal quality factor.

This experiment was planned to be carried out in a Leiden Cryogenics fridge using a setup like the one shown in [18]. It is of interest that the stage where the chip is set to be measured will be at ~ 10 mK.

Moreover, in order to have proper control over the input signal and be able to measure the output properly a setup such as the one shown in [18] was employed. In these schematics several attenuators amounting to 56 dB are installed between the input signal generator and the device; this is motivated by the fact that the power interval desired to work in is as low as the single-photon level; which, in the 4-8 GHz interval, corresponds to -140 dBm to -150 dBm. Furthermore, both infrared and band pass filters are placed right before and after the connection to the device; infrared filters serve as a way to remove infrared radiation from having an effect on the circuit, and band pass filters limit the working range to 4-8 GHz, as well as preventing noise from affecting the measurements.

Additionally, two amplifiers can be seen in the signal chain: one at the 4 K stage, and one at room temperature, right before output acquisition. These amplifiers are crucial to the measurement of the device's response, as the signal reaching the output is inherently low in power and must be amplified in order to carry out the acquisition. This is particularly true when operating at the single-photon level.

In order to study the behaviors discussed in Section 3.3, the measurement was planned to be as follows:

1. A coarse frequency sweep around the respective f_r of each resonator in order to find where the real dip was located.
2. Once the dip was located, a finer sweep, closing in on the f_r and with a higher intermediate frequency (IF) in order to better resolve the spectrum.
3. Once a properly resolved S_{21} spectrum is achieved, exporting the $\Re(S_{21})$ and $\Im(S_{21})$ spectrum.
4. Using the S_{21} method described in [4], extracting the desired data related to Q_i as it is the interest of this thesis; as well as data related to ϵ_{eff}, Z_0 and Q_c in order to compare the results from the fabricated chips with those gotten from the Sonnet simulations.

This process was then planned to be carried out at different temperatures. Starting out at base temperature (~ 10 mK), a series of measurements were to be taken at incrementally higher temperature set points until temperatures of the order of ~ 200 mK, allowing the system to thermalize at each step before recording the resonator response. Around this temperature the frequency shift caused by changes in temperature stop being caused by TLS and quasiparticles become the main contributor[1]. This sweep was to be performed at low, near single-photon power, so that the observed evolution of Q_i and f_r could be attributed to thermal saturation of the TLS population. This way the value of $F\delta^0_{TLS}$ could be extracted following Eq. (2.23) [19].

Additionally, in every step of this measurement plan, several iterations were planned to be carried out on each step at base temperature (~ 10 mK), with increasingly less attenuation. This would then allow us to perform a sweep in power, which is of interest in order to gain further insight into the TLS behavior.

This way, the resulting spectrum obtained from the measurements should look similar to those described in Sections 2.2.2 and 3.3; which means properly fitting the complex S_{21} spectrum to the expression presented by [4].

The real f_r are expected to fall slightly under the predicted values. Also, the experimental values of Q_c are expected to vary significantly, especially at higher values, due to its high sensitivity to any variation in coupling distance.

Due to fabrication and cryostat scheduling constraints, the measurements described in this work were not carried out within the scope of this thesis, although they are expected to be completed before the end of the calendar year. Further work to measure the chips discussed in this text is therefore necessary, as well as future fabrication runs to make up for the unviable chips, including the diamond and 4H-SiC substrates. Discussions were taking place at the time of this work's completion, surrounding the development of a mask for each kind of substrate, in order to carry out runs using photolithography techniques which would facilitate further work.

4 Conclusions and outlook

In this thesis, an in-depth analysis of $\lambda/4$ CPW resonator theory was carried out; encompassing general superconductivity phenomenology, focusing on critical parameters and magnetic flux vortices; and general CPW resonator behavior, with a focus on estimations of ϵ_{eff} and Z_0, as well as a general description of loss channels in the type of circuit studied in this work.

The predictions made possible from the theoretical study were used for the design of a series of chips tailored for the substrates under study. Additionally, experimental design considerations such as the lack of knowledge of the value of Q_i was also taken into account

[1] A more rigorous explanation of the procedure can be found in [19]

in order to achieve critical coupling. Finally, each chip in the ended up consisting in four CPW resonators each with varying f_r and Q_c values.

These resonators were simulated using the method of moments software Sonnet Software. The results of these simulations served as verification of the overall electromagnetic behavior of the circuit, including the current density distribution which proved the resonant state and mode; as well as better estimations for the values of ϵeff, Z_0 and Q_c. The results gotten from these simulations was used to adjust the final parameters of the resonators, which were used for the layouts sent to MPP in Munich, Germany for fabrication.

Once these chips were fabricated, they were assessed for measurement; searching for discontinuities in transmission lines that could cause shorts or open ends which would compromise the measurements. All in all, one 6H-SiC was found to be clearly ready for measurements.

To fulfill the thesis' objectives, the considerations on the setup covered the usage of a cryogenic fridge, capable of arriving at temperatures of $\sim$ 10 mK; and a circuit setup including infrared and LBP filters, as well as TWPA style amplifiers, in order to enable working in the low-photon number regime, as well as increasing the SNR.

Additionally, the considerations on the procedure covered the execution of a temperature sweep as well as a power sweep of Q_i. In this case, the temperature sweep ($Q_i(T)$) serves for the extraction of δ^0_{TLS}; and the power sweep ($Q_i(P)$) served as verification of the saturation of TLS.

Due to fabrication and cryostat scheduling constraints, cryogenic measurements of the measurement-ready 6H-SiC chip could not be carried out in the scope of this thesis. Consequently, the immediate continuation of this work involves executing the outlined frequency, temperature and power sweeps on the 6H-SiC sample to extract its internal quality factor, as well as making a first measurement of 6H-SiC's $\delta^0_T LS$. Future iterations will focus on the optimization of the fabrication process, as well as the development of photolitography masks to facilitate future fabrication runs.

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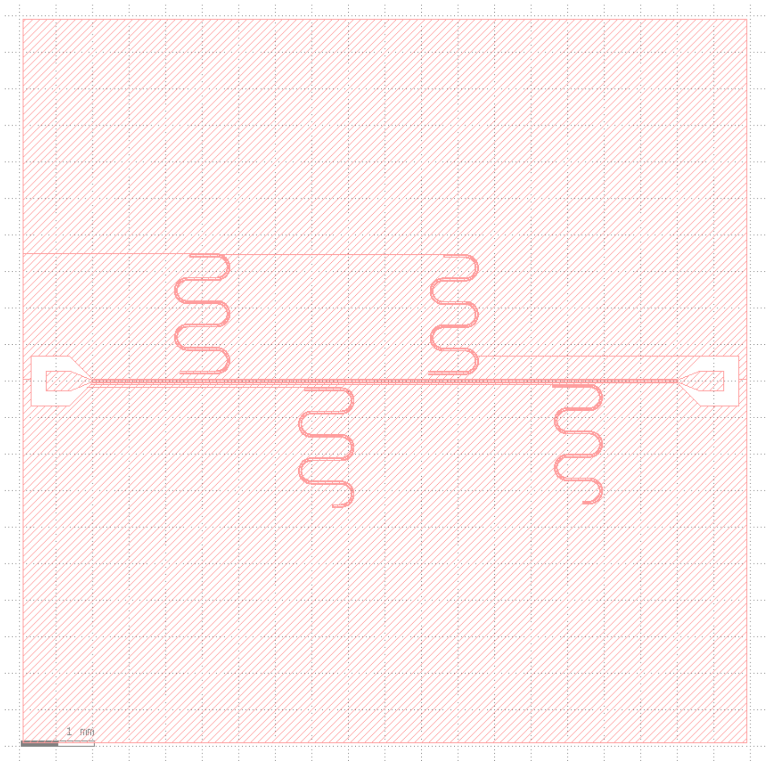
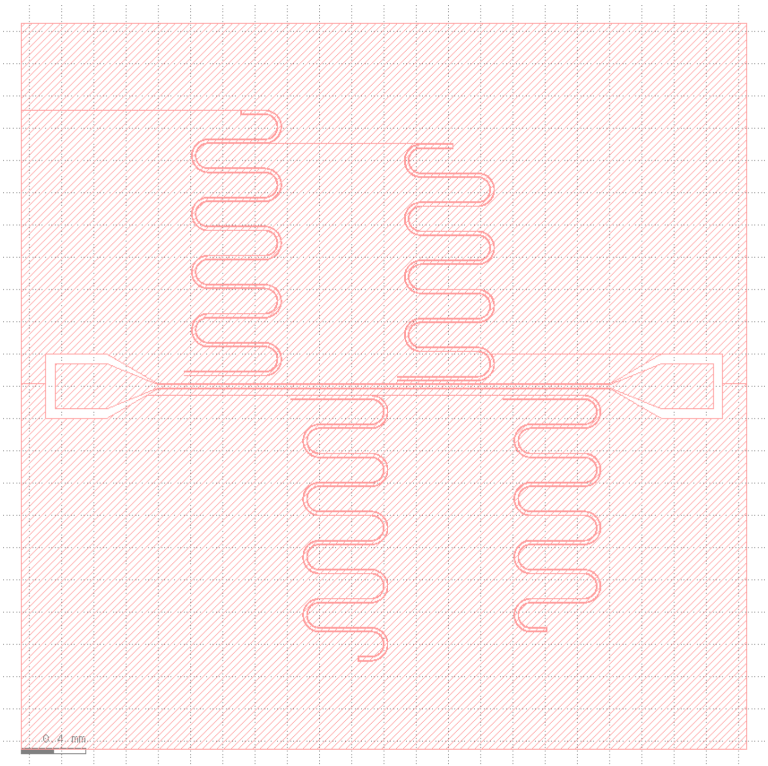
A Resonator chip layouts

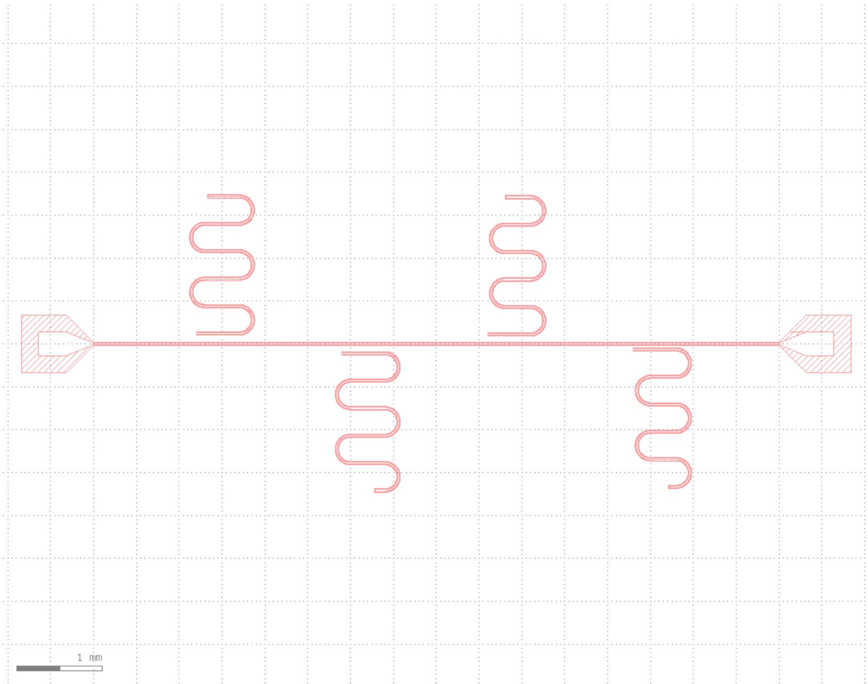


(a) 6H-SiC positive layout

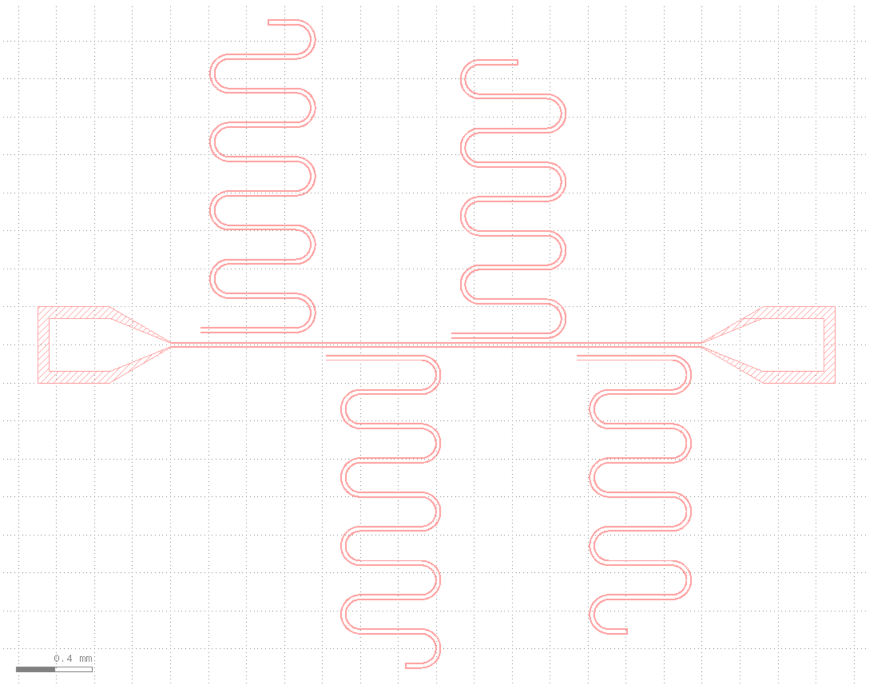


(b) Diamond positive layout

Figure 12: Positive layouts used for the wet etching process.

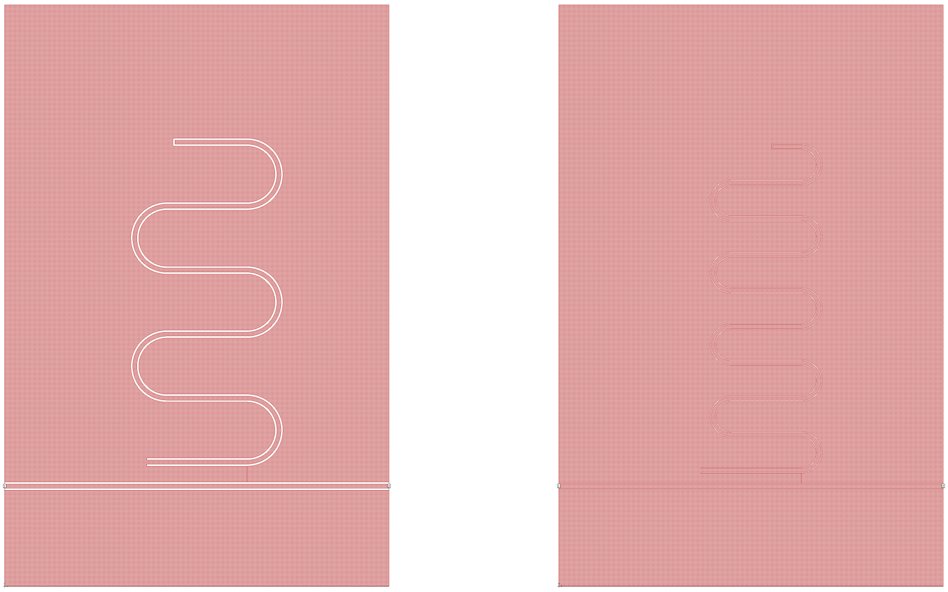


(a) 6H-SiC negative layout



(b) Diamond negative layout

Figure 13: Negative layouts used for the lift-off process.



(a) 6H-SiC simulation layout

(b) Diamond simulation layout

Figure 14: Simulation layouts used for the wet etching process.

The design of the resonator geometries was carried out via a custom Python script built on the `gdstk` library. The script parametrically generates the geometry of each individual $\lambda/4$ CPW resonator (center-conductor width, gap, total length, and coupling geometry to the feedline) from a small set of physical input parameters, allowing the four resonators on each chip to be produced consistently and rapidly regenerated whenever target frequencies or coupling strengths were revised. The same script assembles the individual resonator geometries into two distinct layout variants: a simulation layout, consisting of a single resonator together with its feedline segment and a surrounding ground plane extending $\sim$ 500 µm from the resonator edge on each side, matching the geometry entered into Sonnet; and a fabrication layout, consisting of the complete four-resonator chip with launchers, feedline, and full ground plane, exported as the exposure pattern for the laser writer. Generating both layouts from a single parametric script, rather than drawing them independently, ensures that the simulated and fabricated geometries are identical up to the ground-plane truncation applied in the simulation layout.

The material parameters entered into Sonnet were tailored to the three substrates and to the aluminum film actually used, rather than left at their generic library values.

- **Substrate:** Diamond, being cubic and therefore dielectrically isotropic, was modeled with a single relative permittivity $\varepsilon_{\mathrm{rel}} = 5.7$. For 4H-SiC and 6H-SiC, whose hexagonal polytype structure gives rise to distinct ordinary and extraordinary dielectric axes, anisotropic permittivity values were used instead of a single isotropic figure, with in-plane and out-of-plane components $\varepsilon_{\mathrm{rel},\parallel} = 9.66$ and 9.77, and $\varepsilon_{\mathrm{rel},\perp} =$

10.2 and 10.3 taken from [16, 17], respectively. Substrate thickness was set to 350 μm for both SiC variants and 500 μm for diamond.

- **Metal (Al):** The Surface Impedance loss model was used, appropriate for a superconducting film treated as a 2D boundary condition. The surface resistance was set to $R_s = 0$ to emulate superconductivity, with a film thickness of 300 nm, matching the first-run fabricated samples. No additional sheet inductance value was specified; that is, no manually imposed kinetic inductance correction was applied beyond Sonnet's default surface impedance treatment.

- **Top and box boundaries:** The top layer, representing air above the chip, was assigned $\varepsilon_{\text{rel}} = \mu_{\text{rel}} = 1$ and zero conductivity and loss tangent, with a thickness of 50 000 μm to place the upper boundary far enough from the metal layer to approximate open space. On the contrary, the box bottom was set as a grounded boundary rather than free space, correctly modeling the grounded geometry of the physical chip on a printed circuit board (PCB).

B Resonator fittings

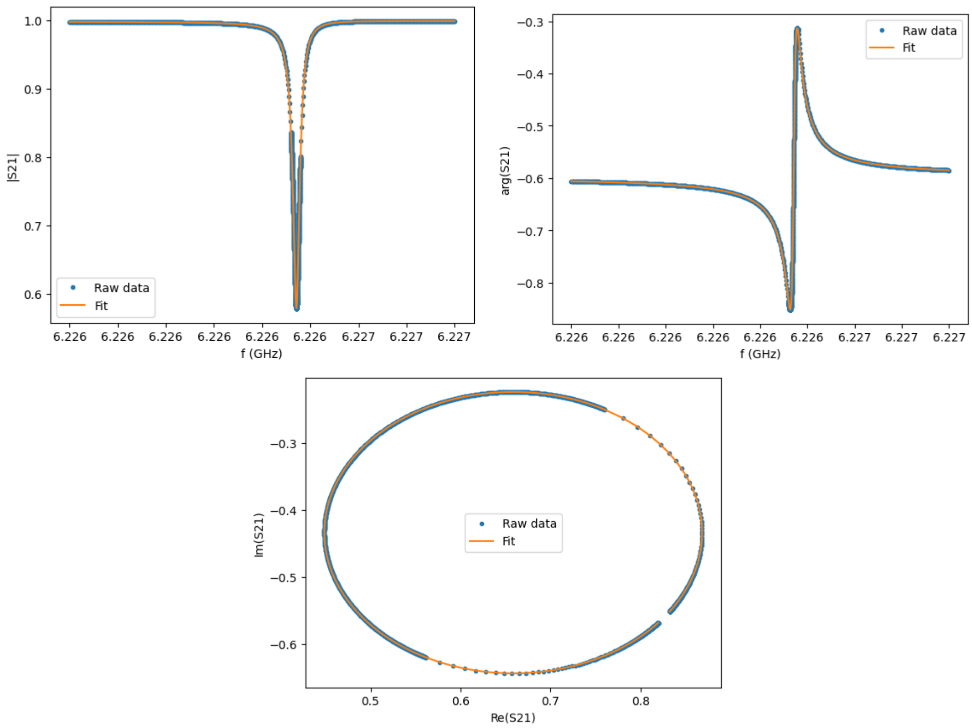


Figure 15: Reproduced from [7]

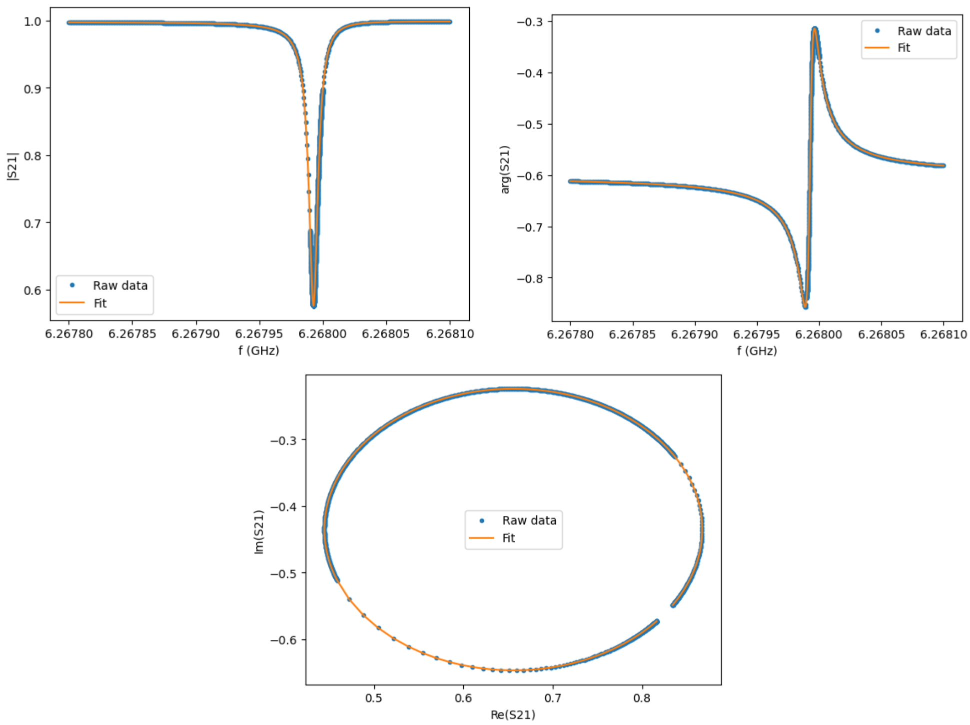


Figure 16: Reproduced from [7]

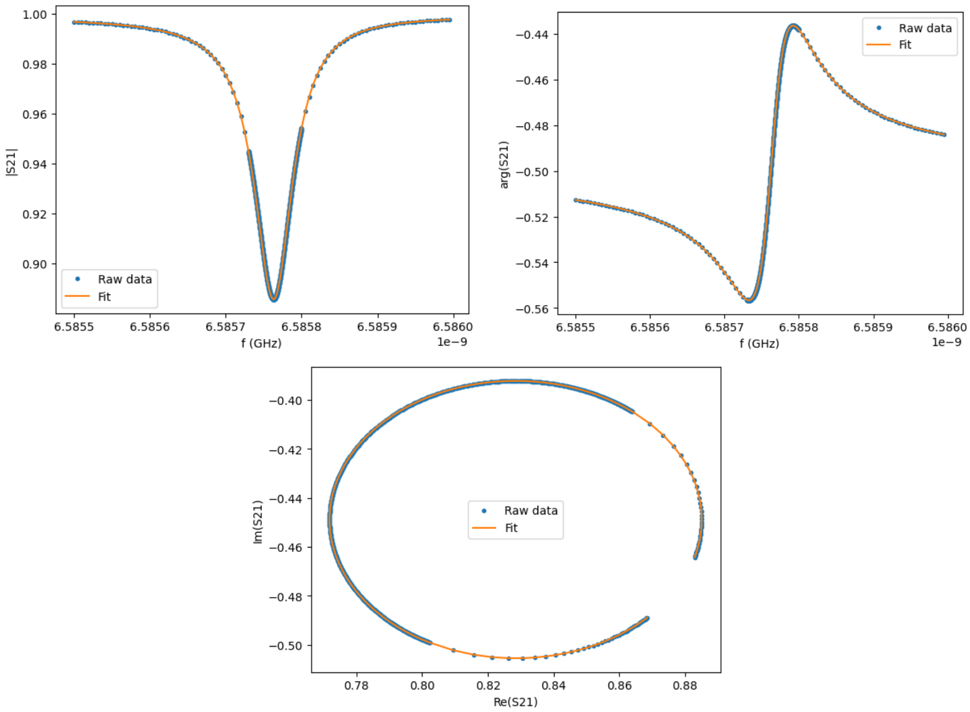
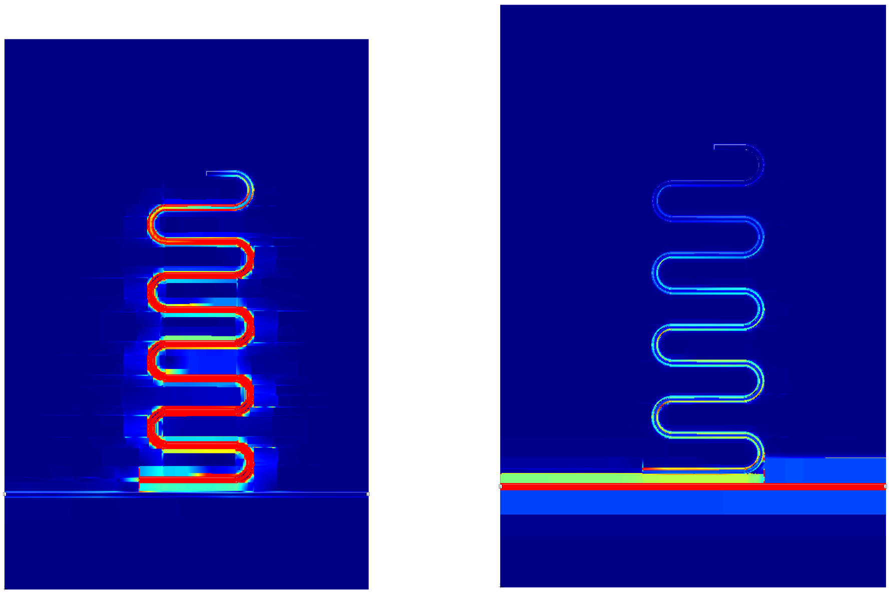


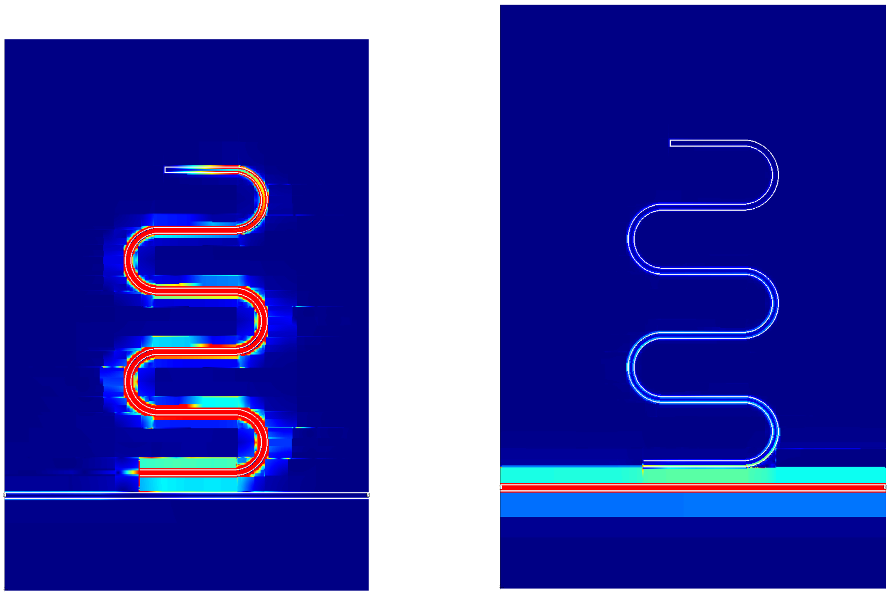
Figure 17: Reproduced from [7]

C Current density distributions



(a) Diamond, in resonance

(b) Diamond, out of resonance



(c) 6H-SiC, in resonance

(d) 6H-SiC, out of resonance

Figure 18: Current density distribution for Diamond and 6H-SiC substrates.